ECA08 - ANALOG AND DIGITAL COMMUNICATION

CO1 DESIGN TASK ASSIGNMENT REPORT
AM-Based Communication Link Design: From Modulation Selection to Receiver Deployment

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STAGE 1: INTERPRET THE COMMUNICATION DESIGN REQUIREMENT (10 Marks)

1. Basic Communication System Block Diagram

A fundamental communication system consists of the following functional blocks:
[Information Source] → [Baseband Signal m(t)] → [Transmitter / Modulator] → [Channel (AWGN Noise)] → [Receiver / Demodulator] → [Recovered Signal m'(t)] → [User Output]
The transmitter processes the message signal onto a high-frequency carrier. The channel introduces noise and attenuation, while the receiver extracts the original audio message.

2. Need for Modulation & Limitations of Direct Baseband Transmission

Direct transmission of unmodulated audio (baseband, 20 Hz – 4 kHz) suffers from severe physical and operational limitations:
• Impractical Antenna Dimensions: Efficient radiation requires antenna length $\lambda \approx \lambda/4$. For f = 4 kHz, $\lambda = c/f = 75$ km, requiring a ~18.75 km antenna. At f_c = 1 MHz, λ = 300 m, making antenna length ~75 m (practically realizable).
• Signal Overlap & Interference: Multiple audio sources transmitting directly in baseband would occupy the exact same frequency spectrum (0–4 kHz), causing unrecoverable mutual interference.
• Frequency Division Multiplexing (FDM): Modulation translates baseband spectra to distinct RF bands, enabling simultaneous broadcasting without interference.

3. Bandwidth Equations & Channel Allocation Capacity

For a maximum voice message frequency f_m = 4 kHz:
• Standard AM: Bandwidth $B_{AM} = 2 \times f_m = 8$ kHz
• DSB-SC: Bandwidth $B_{DSB} = 2 \times f_m = 8$ kHz
• SSB-SC: Bandwidth $B_{SSB} = f_m = 4$ kHz
• VSB-SC: Bandwidth $B_{VSB} = f_m + f_v = 4$ kHz + 1.25 kHz = 5.25 kHz

Channel Allocation in 540 – 1600 kHz Band:
• Total available RF Bandwidth = 1600 kHz - 540 kHz = 1060 kHz.

• Given Channel Spacing = 20 kHz.
• Maximum Idealized Channels = Total Bandwidth / Channel Spacing = 1060 / 20 = 53 channels.

4. Power Calculations (P_c = 10 W, m = 0.7)

Given Carrier Power P_c = 10 W and modulation index m = 0.7:
• Sideband Power: P_SB = (m^2 / 2) * P_c = (0.7^2 / 2) * 10 = 0.49 / 2 * 10 = 2.45 W
• Upper Sideband Power (P_USB) = Lower Sideband Power (P_LSB) = P_SB / 2 = 1.225 W
• Total Transmitted Power: P_T = P_c * (1 + m^2 / 2) = 10 * (1 + 0.245) = 12.45 W
• Power Efficiency: η = (P_SB / P_T) * 100% = (2.45 / 12.45) * 100% = 19.68%

> ■ **STAGE 1 DELIVERABLE: DESIGN REQUIREMENT STATEMENT**
>
> Message Bandwidth (f_m): 4 kHz | RF Spectrum: 540 - 1600 kHz | Carrier (f_c): 1 MHz | Spacing: 20 kHz
>
> Max 20-kHz Channels: 53 | Total Transmitted Power: 12.45 W (P_c = 10 W, P_SB = 2.45 W, η = 19.68%)
>
> Key Constraints: Educational low-cost prototype, envelope detection compatibility, AWGN tolerance.

STAGE 2: INVESTIGATE AND SELECT THE MODULATION SCHEME (20 Marks)

1. Modulation Scheme Comparison Table

Scheme	BW (f_m=4kHz)	Carrier	Power Efficiency	Complexity	Typical Application
AM	8 kHz	Transmitted	Moderate (~19.7%)	Low (Simple envelope detector)	AM Broadcast Radio
DSB-SC	8 kHz	Suppressed	High (100% sidebands)	Moderate (Coherent carrier sync)	Point-to-point links
SSB-SC	4 kHz	Suppressed	High (100% sideband)	High (Sharp filters / Hilbert)	HF Voice, Amateur Radio

VSB-SC	5.25 kHz	Controlled	High	High (Asymmetric filtering)	Analog TV Video Signal

2. Mathematical Derivations

Let $m(t) = A_m \cos(2\pi f_m t)$ and carrier $c(t) = A_c \cos(2\pi f_c t)$.
• Standard AM: $s_{AM}(t) = A_c [1 + m \cos(2\pi f_m t)] \cos(2\pi f_c t) = A_c \cos(2\pi f_c t) + (m A_c / 2) \cos(2\pi (f_c + f_m) t) + (m A_c / 2) \cos(2\pi (f_c - f_m) t)$
• DSB-SC: $s_{DSB}(t) = m(t) c(t) = A_m A_c \cos(2\pi f_m t) \cos(2\pi f_c t) = (A_m A_c / 2) [\cos(2\pi (f_c + f_m) t) + \cos(2\pi (f_c - f_m) t)]$
• SSB-SC (USB): $s_{SSB}(t) = (A_m A_c / 2) \cos(2\pi (f_c + f_m) t) = (1/2) m(t) \cos(2\pi f_c t) - (1/2) \hat{m}(t) \sin(2\pi f_c t)$, where $\hat{m}(t)$ is the Hilbert transform.
• VSB-SC: $s_{VSB}(t) = [m(t) \cos(2\pi f_c t)] * h_v(t)$, where $h_v(t)$ is the impulse response of a vestigial sideband filter.

3. Rationale for VSB in Analog Television

Analog television signals contain video baseband frequencies up to 4.2 MHz. Transmitting DSB-SC would require 8.4 MHz bandwidth per channel. While SSB-SC would save bandwidth (4.2 MHz), video signals contain significant low-frequency DC components, making pure SSB filter design practically impossible due to the sharp cutoff required at DC. VSB passes one full sideband and a small 'vestige' (1.25 MHz) of the other, simplifying filter design while saving ~3 MHz compared to DSB.

■ DESIGN DECISION 1: MODULATION SELECTION

RECOMMENDATION: Standard Amplitude Modulation (AM) with m = 0.7.

JUSTIFICATION: Although DSB-SC and SSB-SC offer superior power efficiency (100% vs 19.7%), Standard AM allows the use of an extremely simple, low-cost passive envelope detector (diode + RC circuit) at the receiver.

For an educational prototype community broadcast link, suppressing the carrier would require complex phase-locked loops (PLL) or Costas loops for coherent detection, drastically increasing receiver cost and complexity. AM satisfies the 8 kHz bandwidth limit within the allocated 20 kHz channel spacing.

STAGE 3: DEVELOP RF CHANNEL ALLOCATION AND TRANSMITTER DESIGN (15 Marks)

1. Representative RF Channel Plan (Around 1 MHz)

Channel	Carrier (f_c)	Occupied Spectrum	Guard Band	Status
Ch 23	960 kHz	956 - 964 kHz	6 kHz	Adjacent Channel
Ch 24	980 kHz	976 - 984 kHz	6 kHz	Adjacent Channel
Ch 25 (Assigned)	1000 kHz (1 MHz)	996 - 1004 kHz	6 kHz	Design Channel
Ch 26	1020 kHz	1016 - 1024 kHz	6 kHz	Adjacent Channel
Ch 27	1040 kHz	1036 - 1044 kHz	6 kHz	Adjacent Channel

2. Spectral Occupancy & Guard Band Calculations

• Channel Center Frequency: f_c = 1000 kHz (1 MHz)
• Signal Occupied Bandwidth: B_AM = 2 × 4 kHz = 8 kHz (from 996 kHz to 1004 kHz)
• Allocated Channel Spacing: 20 kHz (from 990 kHz to 1010 kHz)
• Guard Band = Channel Spacing - Occupied Bandwidth = 20 kHz - 8 kHz = 12 kHz total (6 kHz on each side).
• Spectral Overlap: Zero spectral overlap exists between adjacent channels due to the generous 6 kHz guard bands.

3. Transmitter Block Diagram Architecture

The low-level transmitter architecture consists of:
[Microphone / Audio Source] → [Baseband LPF (4 kHz Cutoff)] → [Audio Preamplifier] → [AM Collector Modulator (1 MHz Carrier from Crystal Oscillator)] → [Class-C RF Power Amplifier (10 W Carrier)] → [Impedance Matching Network] → [Antenna]

■ **DESIGN DECISION 2: TRANSMITTER ARCHITECTURE & SPECTRUM ALLOCATION**

RECOMMENDATION: High-Level AM Collector Modulator with Crystal Oscillator Carrier Generation at 1000 kHz.

JUSTIFICATION: Offers highly stable carrier generation, excellent power efficiency in the RF power stage, and simple integration for educational laboratory demonstration without complex filter tuning required by SSB/VSB.

STAGE 4: SELECT THE AM RECEIVER ARCHITECTURE (20 Marks)

1. Receiver Comparison: TRF vs. Superheterodyne

Parameter	Tuned Radio Frequency (TRF)	Superheterodyne Receiver
Sensitivity	Varies with tuned frequency; poor at high RF	High and constant across entire band
Selectivity	Degrades at higher frequencies (Q = f_c / BW)	High and uniform (determined by fixed IF filter)
Fidelity	Poor due to varying bandwidth across band	Excellent and consistent audio response
Stability	Prone to multi-stage oscillation/instability	Highly stable due to fixed IF amplification
Image Rejection	Not applicable (no mixer used)	Requires adequate RF front-end filtering
Complexity	Low circuit count, difficult multi-stage tuning	Moderate (Mixer, Local Oscillator, IF filter)

2. Superheterodyne Block Diagram & Calculations

Superheterodyne Receiver Functional Flow:
[RF Antenna] → [RF Amplifier / BPF] → [Mixer] ← [Local Oscillator (f_LO)] → [IF Amplifier (f_IF = 455 kHz)] → [Envelope Detector] → [AGC Loop & Audio Amplifier] → [Speaker]

Frequency Design Calculations (for f_c = 1000 kHz, standard IF f_IF = 455 kHz):
• Local Oscillator Frequency: f_LO = f_c + f_IF = 1000 kHz + 455 kHz = 1455 kHz
• Image Frequency: f_img = f_c + 2 × f_IF = 1000 kHz + 2 × 455 kHz = 1910 kHz
• Influence of IF on Image Rejection: Higher IF increases the separation between desired RF and image frequency (2 × f_IF), improving RF preselector rejection. 455 kHz provides an optimal trade-off between image rejection and adjacent channel selectivity.

> **■ DESIGN DECISION 3: RECEIVER ARCHITECTURE**
>
> RECOMMENDATION: Superheterodyne Receiver Architecture with IF f_IF = 455 kHz and Envelope Detection.
>
> JUSTIFICATION: The superheterodyne structure provides uniform selectivity and constant high sensitivity across the 540-1600 kHz band by converting all incoming RF signals to a fixed 455 kHz IF. Image rejection at 1910 kHz is easily achieved using a standard single-stage RF preselector.

STAGE 5: DEVELOP, VERIFY AND COMPARE THE COMPLETE SYSTEM (15 Marks)

1. End-to-End System Spectrum Analysis

• Point A (Baseband): Audio voice spectrum spanning 0 – 4 kHz.
• Point B (Transmitted AM RF): Carrier at 1000 kHz with LSB (996–1000 kHz) and USB (1000–1004 kHz).
• Point C (Mixer Output): Intermediate frequency centered at 455 kHz with sidebands from 451–459 kHz.
• Point D (Recovered Audio): Demodulated voice audio baseband (0 – 4 kHz) with low distortion.

2. Comparative Performance with Rejected Alternative (DSB-SC)

• Standard AM (Selected): Requires 12.45 W total power, 8 kHz bandwidth. Demodulation requires only 1 diode and 1 RC circuit costing < $0.50.
• DSB-SC (Rejected): Requires only 2.45 W total power (10 W savings), 8 kHz bandwidth. However, demodulation requires carrier synchronization (PLL/Costas loop) with phase error sensitivity. A 10°

carrier phase error degrades recovered audio signal-to-noise ratio by > 1.2 dB and adds prohibitive prototype circuit complexity.

STAGE 6: CRITICALLY EVALUATE THE DESIGN AND MAKE THE DEPLOYMENT DECISION (20 Marks)

1. Specification Verification Matrix

Parameter	Required Specification	Designed System Value	Compliance Status
Message Bandwidth	4 kHz max audio	4 kHz LPF applied	COMPLIANT
Carrier Frequency	1 MHz (540-1600 kHz)	1000 kHz (Ch 25)	COMPLIANT
Channel Spacing	20 kHz spacing	20 kHz (6 kHz guard band)	COMPLIANT
Modulation Index	m = 0.7	m = 0.7 (19.7% efficiency)	COMPLIANT
Transmitter Power	10 W Carrier Power	P_c = 10 W, P_T = 12.45 W	COMPLIANT
Receiver Architecture	Educational Prototype	Superheterodyne (455 kHz IF)	COMPLIANT

2. Comparative Suitability: Voice Broadcast vs. Analog TV Transmission

• Voice Broadcasting: Standard AM is ideal due to low audio bandwidth (4 kHz) and the imperative for cheap, mass-producible receiver units (envelope detectors).
• Analog TV Transmission: Standard AM is highly unsuitable for TV video due to huge bandwidth consumption (8.4 MHz for DSB vs 5.25 MHz for VSB). VSB is mandatory for TV because it accommodates DC video content while conserving ~3 MHz of precious spectrum per channel.

■ FINAL ENGINEERING DEPLOYMENT DECISION

DEPLOYMENT DECISION: YES (DEPLOY SYSTEM)

DEFENSE OF DECISION:

1. Full Specification Compliance: The proposed system fully satisfies all voice audio bandwidth (4 kHz), power (10 W carrier, 12.45 W total), and RF spectrum allocation constraints (1 MHz carrier, 20 kHz spacing).

2. Receiver Simplicity & Cost: Utilizing Standard AM with m = 0.7 allows simple envelope detection in a Superheterodyne architecture (f_{IF} = 455 kHz), ideal for an educational/community prototype.

3. Spectral Isolation: Occupies 8 kHz out of 20 kHz allocated spacing, providing 6 kHz guard bands on both sides to prevent adjacent channel interference.

REDESIGN CONTINGENCY PLAN (If spectrum congestion increases):

If channel spacing is reduced from 20 kHz to 10 kHz or power efficiency becomes critical, switch to SSB-SC with a 455 kHz mechanical filter and a synchronous product detector. This would halve bandwidth consumption to 4 kHz and reduce transmitted power to 2.45 W, at the cost of higher receiver circuit complexity.